

Computational Biomathematics: Project Portfolio

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The following project descriptions are designed to guide you into complex biological systems where interactions between agents and their environment give rise to rich and often unexpected dynamics. Each case is presented as a narrative to encourage you to think like a modeler: identifying relevant agents, interactions, environmental fields, and feedback loops.

You are expected to translate these biological stories into computational models, implement them in a modular way, simulate their behavior, and interpret the results scientifically. For each project, a set of required outputs is provided. These define the minimum expectations for successful completion.

All projects follow a structured progression from simple to complex models. Students are expected to iteratively extend their models and validate each stage before proceeding.

Step A (Minimal Model) introduces the core agents and basic interactions without environmental coupling. The focus is on correct implementation and baseline dynamics.

Step B (Biological Mechanisms) enriches the model with biologically meaningful processes such as lifecycle stages, behavioral rules, or interaction specificity.

Step C (Environmental Fields) introduces spatially distributed fields such as nutrients, signals, or environmental gradients and couples them to agent dynamics.

Step D (Full System) integrates feedback loops, adaptation, interventions, or evolutionary effects, leading to a fully coupled ecosystem model.

Students are required to use AI tools to support the development of their simulation code. This includes generating modular components such as movement rules, interaction functions, and field updates. Students must document their prompts and critically evaluate the generated code.

Testing must be performed using `pytest`, with students designing their own test cases for correctness, stability, and boundary conditions.

1. Tumor–Immune–Microenvironment Ecosystem

A tumor does not grow in isolation but exists within a highly dynamic microenvironment. Cancer cells consume oxygen and nutrients, altering the local conditions in such a way that regions of hypoxia emerge. These hypoxic zones often give rise to more aggressive and therapy-resistant tumor phenotypes. At the same time, immune cells infiltrate the tumor, attempting to eliminate malignant cells. However, tumors actively suppress immune responses by releasing signaling molecules that inhibit immune activity or attract regulatory immune cells that dampen the attack.

In this system, the spatial structure is crucial. Regions of high tumor density may exclude immune cells, while nutrient gradients shape the direction of tumor expansion. Therapeutic interventions, such as drugs or immunotherapy, further modify the environment and can lead to highly non-linear outcomes depending on timing, dosage, and spatial distribution.

Students are encouraged to explore how local interactions between tumor cells, immune cells, and environmental fields lead to global patterns such as tumor persistence, collapse, or oscillatory behavior.

Project structure

Step A. Students construct a minimal model of tumor growth with simple proliferation and death rules, alongside immune cells that randomly encounter and eliminate tumor cells.

Step B. The system is extended by introducing tumor heterogeneity (e.g., proliferative vs. hypoxic states) and differentiated immune responses.

Step C. A spatial oxygen or nutrient field is introduced, influencing tumor growth and immune activity. Diffusion and consumption must be modeled explicitly.

Step D. Feedback mechanisms are incorporated, such as immune suppression by tumor signaling or drug treatment fields, allowing exploration of therapy strategies.

Must-have results

- Time series of tumor cell populations (by phenotype if modeled)
- Time series of immune cell populations
- Spatial heatmap of oxygen or nutrient field
- Spatial distribution of tumor and immune cells over time
- Quantification of tumor growth rate under different conditions
- Effect of introducing therapy (e.g., drug field) on tumor dynamics
- Visualization of hypoxic regions and their evolution

2. Biofilm Formation under Flow and Antibiotic Stress

Bacteria in natural environments often transition from a free-swimming (planktonic) state to a surface-attached community known as a biofilm. Within these biofilms, bacteria produce extracellular matrix substances that protect them from environmental stress, including antibiotics. The formation of a biofilm is often triggered by quorum sensing signals, which allow bacteria to detect local population density.

In environments with fluid flow, nutrients and antibiotics are transported dynamically, leading to spatial heterogeneity. Some regions of the biofilm may receive sufficient nutrients but little antibiotic exposure, while others experience the opposite. This spatial shielding leads to the emergence of highly resistant subpopulations, including so-called persister cells.

Students should investigate how flow, signaling, and spatial structure combine to produce emergent resistance and long-term survival of bacterial populations.

Project structure

- Step A.** Students model bacteria that switch between planktonic and attached states with simple growth dynamics.
- Step B.** Quorum sensing is introduced, allowing bacteria to detect local density and initiate biofilm formation.
- Step C.** Nutrient and antibiotic fields are added, including spatial diffusion and consumption effects.
- Step D.** A flow field is implemented, transporting nutrients and antibiotics and shaping spatial heterogeneity in the system.

Must-have results

- Time series of planktonic vs. attached bacteria
- Spatial visualization of biofilm growth
- Heatmap of nutrient concentration
- Heatmap of antibiotic concentration
- Visualization of quorum sensing signal field
- Comparison of bacterial survival with and without biofilm formation
- Analysis of spatial heterogeneity in resistance

3. Coral Reef Collapse and Recovery

Coral reefs are delicate ecosystems where coral organisms, algae, and herbivorous fish interact in a tightly coupled system. Under normal conditions, corals dominate and provide structure to the ecosystem. However, environmental stress such as increased temperature can cause coral bleaching, weakening the coral and allowing algae to overgrow.

Once algae dominate, recovery becomes difficult unless herbivorous fish populations are sufficient to reduce algal coverage. Nutrient levels further complicate this balance, as high nutrient availability can favor algal growth.

This project explores tipping points and regime shifts, where small changes in environmental conditions can lead to large and potentially irreversible changes in ecosystem structure.

Project structure

Step A. A simple competition model between coral and algae is implemented.

Step B. Herbivorous fish are introduced as agents that reduce algae.

Step C. Environmental fields such as temperature and nutrients are incorporated.

Step D. Feedback loops such as bleaching and recovery dynamics are added, enabling tipping point analysis.

Must-have results

- Time series of coral, algae, and fish populations
- Spatial maps of coral vs. algae dominance
- Visualization of temperature field effects
- Analysis of bleaching events and recovery
- Identification of tipping points in parameter space
- Effect of varying fish populations on system stability
- Phase diagram showing regimes (coral-dominated vs. algae-dominated)

4. Predator–Prey Dynamics with Learning and Fear

In natural ecosystems, prey do not simply move randomly but learn to avoid areas where predators are frequently encountered. Similarly, predators adapt their hunting strategies based on past success. This leads to the emergence of spatial memory and behavioral adaptation.

Over time, prey may avoid certain regions entirely, creating de facto safe zones, while predators concentrate their activity in high-success areas. These feedback loops fundamentally alter classical predator–prey dynamics and can lead to unexpected outcomes such as spatial segregation or oscillatory patterns.

Students should explore how simple learning rules and memory effects can reshape ecological interactions.

Project structure

Step A. Classical predator–prey interaction is implemented with random movement.

Step B. Behavioral adaptation is introduced, allowing predators and prey to adjust movement.

Step C. A memory or fear field is added, influencing prey movement.

Step D. Learning mechanisms (e.g., reinforcement-like rules) are integrated.

Must-have results

- Time series of predator and prey populations
- Spatial distribution of prey over time
- Visualization of “fear field” or memory field
- Predator movement patterns and clustering
- Comparison with non-learning baseline model
- Analysis of prey survival rates under learning
- Identification of emergent spatial patterns

5. Wound Healing under Infection Pressure

When tissue is damaged, a complex process of wound healing begins, involving immune cells, repair cells, and signaling molecules. However, if bacteria infect the wound, the healing process becomes more complicated. Immune responses must balance eliminating pathogens with avoiding excessive tissue damage.

Chemical signals guide cells toward the wound site, while oxygen and nutrient availability determine the speed of regeneration. Infection introduces competing dynamics, where bacterial growth may outpace immune clearance.

Students should investigate how healing and infection compete and under what conditions recovery is successful.

Project structure

Step A. Tissue recovery dynamics are modeled without infection.

Step B. Immune response and infection agents are introduced.

Step C. Signaling fields guide immune and repair cells.

Step D. Competing dynamics between healing and infection are analyzed under varying conditions.

Must-have results

- Time series of tissue recovery (e.g., damaged vs. healed cells)
- Time series of bacterial population
- Spatial visualization of wound closure
- Heatmap of signaling or growth factor field
- Effect of infection on healing time
- Comparison of immune response strengths
- Identification of conditions leading to chronic wounds

6. Invasive Species and Habitat Engineering

Some invasive species do not merely compete with native species but actively alter their environment in ways that favor their own survival. For instance, they may change soil composition, water flow, or vegetation structure. These changes often disadvantage native species and create a feedback loop that accelerates the invasion process.

This project focuses on how environmental modification by one species can reshape the competitive landscape and lead to large-scale ecological transformation.

Project structure

Step A. Native and invasive species compete for resources.

Step B. Differences in reproduction and survival strategies are added.

Step C. Environmental fields representing habitat quality are introduced.

Step D. Feedback loops where invasive species modify the environment are implemented.

Must-have results

- Time series of native and invasive populations
- Spatial maps of habitat quality
- Visualization of environmental modification over time
- Analysis of invasion speed
- Comparison with non-engineering invasive species
- Identification of stable vs. unstable coexistence
- Sensitivity analysis with respect to environmental feedback strength

7. Neural Development and Synaptic Pruning

During brain development, neurons form connections that are later refined through a process known as synaptic pruning. Connections that are frequently used are strengthened, while others are eliminated. This process is influenced by neural activity and chemical signaling.

The system evolves dynamically, with structure and function influencing each other. Over time, highly efficient network structures emerge from initially redundant connections.

Students should model how activity-dependent mechanisms lead to network formation and optimization.

Project structure

Step A. A network of neurons with random connections is created.

Step B. Synaptic strengthening and weakening rules are introduced.

Step C. Activity or signaling fields influence connection dynamics.

Step D. Pruning mechanisms and adaptive rewiring lead to emergent structure.

Must-have results

- Evolution of network connectivity over time
- Visualization of network structure (graph representation)
- Distribution of synaptic strengths
- Analysis of pruning dynamics
- Comparison of activity-driven vs. random pruning
- Emergence of clusters or functional modules
- Metrics such as average degree or path length over time

8. Collective Migration with Leaders and Followers

In many biological systems, groups of organisms move collectively without centralized control. Some individuals act as leaders, sensing environmental gradients, while others follow by aligning their movement with neighbors.

Obstacles, crowding, and environmental signals influence movement patterns, leading to complex flow-like behavior. Small changes in leader behavior can significantly alter the trajectory of the entire group.

Students should explore how local interaction rules lead to coordinated global motion.

Project structure

Step A. Agents move randomly in space.

Step B. Alignment and cohesion rules are introduced.

Step C. External fields such as chemoattractants guide movement.

Step D. Leader–follower dynamics and obstacle interactions are implemented.

Must-have results

- Trajectories of agents over time
- Visualization of group movement patterns
- Heatmap of chemoattractant field
- Comparison of leader vs. no-leader scenarios
- Quantification of group cohesion
- Effect of obstacles on migration paths
- Measurement of alignment or order parameter

9. Rhizosphere Ecosystem: Roots, Microbes, and Soil

Plant roots interact with a diverse community of microorganisms in the soil, forming a complex micro-ecosystem known as the rhizosphere. Roots release chemical exudates that attract beneficial microbes, which in turn enhance nutrient availability. At the same time, pathogenic organisms compete for the same space and resources.

The interplay between roots, microbes, and soil chemistry leads to dynamic feedback loops that influence plant growth and health. Spatial structure plays a key role, as interactions occur locally but propagate through diffusion and movement.

Students should investigate how these interactions shape the development of the system over time.

Project structure

Step A. Root growth is modeled in a static nutrient field.

Step B. Microbial populations are introduced.

Step C. Chemical signaling and nutrient feedback are added.

Step D. Full ecosystem coupling with competing microbial species is implemented.

Must-have results

- Growth dynamics of plant roots
- Population dynamics of beneficial and harmful microbes
- Spatial distribution of nutrient field
- Visualization of root exudate field
- Interaction patterns between microbes and roots
- Effect of microbial composition on plant growth
- Identification of stable vs. unstable ecosystem configurations

Grading Rubric (100 Points Total)

Intermediate Result Presentation (30 Points)

The intermediate result presentation (seeing that you are on track) will account for 30% of the grade. It is scheduled for the end of the course (Friday 2nd week) on-site WCU. For details, please see the lecture slides. For each group, the intermediate result presentation is 15 min with additional questions and answers. Criteria are well-defined problem and solution approach as well as first indicative results.

Clear Presentation (10 Points) Clarity of the presentation incl. peer contribution statement, orthography and grammar as well as soundness of the argumentation stream.

Clear Problem Statement and Solution Approach (15 Points) Clarity and transparency of the problem statement (show that you have understood the problem). Clarity and transparency of your solution approach, i.e., describe your solution approach clearly and transparently and shortly illustrate the benefits of this approach.

Consider and carry-out several ways to solve the problem by different approaches:

- Besides your approach, state alternative approaches (at least two) how the problem under consideration can be solved as well
- Summarize, whether (where and how) these approaches were already used (literature research)
- Briefly discuss the pros and cons of the approaches
- Reason, why your solutions approach is better suited

Project Roadmap (5 Points) Clear project roadmap (show that you know what to do in order to solve the problem), i.e., for instance time frame or any other way of presenting the logical order by which you aim to execute the solution approach.

Final Presentation (70 Points)

The final presentation (= report) will account for 70% of the grade and take place virtually about three weeks after the end of the course. For each group, the final result presentation is 30 min with additional questions and answers. Criteria comprise a clear presentation and comprehensive results.

Clear Presentation (15 Points) Clarity of the presentation incl. peer contribution statement, orthography and grammar as well as soundness of the argumentation stream.

Model Design (10 Points) Clarity and appropriateness of the chosen modeling approach, including abstraction of biological processes and justification of assumptions.

Implementation Quality (10 Points) Code structure, modularity, readability, and correctness. Proper use of a structured project architecture.

AI-Assisted Development (5 Points) Quality of prompts, integration of AI-generated code, and critical reflection on AI usage.

Testing and Validation (5 Points) Use of test cases, validation of model behavior, and robustness checks.

Simulation Results (10 Points) Completeness and correctness of required outputs, including plots, spatial visualizations, and animations.

Analysis and Interpretation (10 Points) Depth of insight into the model behavior, including explanation of observed phenomena and parameter sensitivity.

Creativity and Extensions (5 Points) Additional features, novel ideas, or particularly insightful extensions.